Literal certificate: explicit ring normalization

Verification v3 · 2026-10-25 · terminal allocation failure

The same sealed p37 seed0 phi system-0 certificate is exported after a passing 13-test native gate. Source freeze bc818088c1af precedes the unique 07:10:27 UTC managed launch, PID 3851339. Only the instance proof tactics change to explicit grobner; all coefficients, balanced sums, four product lemmas, three final statements and ten axiom audits remain required.

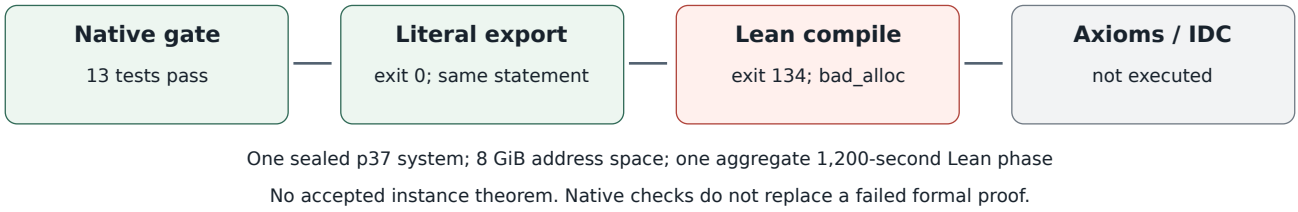

Figure 1: Actual terminal verification gates. No theorem, axiom audit or IDC is inferred from a passing export.

Actual outcome and resource evidence

The export exits 0 and retains the same exact integer-lift statement. All 2,197 dense coefficient positions and the 454 nonzero R coefficients are retained. The intended identity is $\sum h_i f_i = q^2 + 37R$; its complete literal commutative-ring proof and characteristic-37 consequence remain open. Statement SHA-256: `e7dd6d7622171f18c1d80ea9c1d26d182b08dbf664c3a3046115eb6715e5eb45`

The Lean phase starts at 07:20:28 UTC and terminates at 07:32:28 UTC with exit 134, signal 6: libc++abi reports an uncaught std::bad_alloc. Maximum recorded RSS is 3,772,392 KiB under the unchanged 8 GiB address-space bound. These are different memory measures. The one aggregate 1,200-second bound was not increased. Shared-host durations are diagnostic and unadmitted.

GNU time reports signal 6 and then an “Exit status: 0” footer. That footer does not indicate success. The actual phase and terminal receipts both record 134. The driver stops before the axiom audit and IDC. No accepted instance theorem comes from this run or either earlier memory-failed encoding.

Exact remaining proof obligation

Prove the literal integer identity in a commutative ring, then its characteristic-37 q^2 consequence and the repeated-abscissa conclusion under the explicit no-zero-divisor hypothesis. No certificate identity may be assumed. Permit only propext, Classical.choice and Quot.sound in all ten axiom sets; no sorry, custom axiom, native evaluation or kernel bypass. Lean is pinned to 4.34.1; lean4checker was not available.

These failures do not disprove the original finite-field certificate and do not locate the memory failure within elaboration or normalization. Before another large attempt, independently verify the serialized integer statement and isolate small definition/product blocks. Preserve the full statement, explicit hypotheses and aggregate execution bound.

Scope and retained artifacts

No relation collection, matrix rank, target recovery, new complete-cost comparison or general group/family theorem is established here. The native parser, field/group/decoder bridges remain separate. The original 189+21 study and completed twelve-configuration diagnostic are not rerun.

See PROOF-V3.md, native/src/main.rs, PROTOCOL.md and the source-pinned run driver. Raw parent runs/: certificate-instance-native-gate-v3, certificate-instance-proof-v3 and certificate-instance-v3-layout-v1/v2. Compact failed receipts remain under results/certificate-instance-proof-v3/. The deterministic runs-v3.tar.xz and checksum preserve this run without replacing the v1/v2 archive. All actual PDF pages are inspected. No canonical IC/rho graph, leaderboard or benchmark index is promoted by this failure.